1) HARD-CODED INITIANZATION (AND GATE) FIXED VALUES

```
import numpy as np
# Input data
x1= np.array([0,0,1,1])
x2= np.array([0,1,0,1])
t = np.array([0,0,0,1])
m= len(t)
max_iter = 4000
learn_rate = 0.1
# hard coded weights and bias
w1= 0.1
w2= 0.1
b = 0.0
# sigmoid function
def sigmoid(z):
    return 1/(1+np.exp(-z))
# sigmoid derivative
def sigmoid_derivative(s):
    return s*(1-s)
# Training loop
for _ in range(max_iter):
    dw1=0
    dw2=0
    db =0
    for i in range(m):
        z= x1[i] * w1 + x2[i]*w2 + b
        y= sigmoid(z)
        error= t[i]-y
        dw1 += error * sigmoid_derivative(y)* x1[i]
        dw2 += error * sigmoid_derivative(y) * x2[i]
        db += error * sigmoid_derivative(y)
    w1 += learn_rate * (dw1/m)
    w2+ = learn_rate * (dw2/m)
    b += learn_rate *(db/m)
# Output weights and bias
print("w1:", w1, "w2:", w2, "b:", b)
# final prediction
final_z =x1 * w1 + x2 * w2 + b
prediction = sigmoid(final_z)
print("prediction:", prediction.round())
```

OUTPUT

```
w1: 2.6193566678513
w2: 2.6193566678513
b : -4.0547181028
prediction: [0. 0. 0. 1
```